

David Kenechukwu Okonkwo

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EDUCATION

African Leadership University, Rwanda

June 2026

Bachelor of Science in Software Engineering

Cumulative GPA: 4.56/5.0

Relevant Coursework: Enterprise Web Development, Mobile App Development, Data Structures and Algorithms, Low-level Programming, Blockchain and Applications, Embedded Systems

EXPERIENCE

Lead Frontend Developer, Talk Wide – Enugu, Nigeria

November 2024 - Present

- **Led the design and development** of a responsive, scalable student marketplace platform using **React, TypeScript, and Tailwind CSS**, improving user engagement by **35%** and optimizing navigation with **TanStack Router**.
- **Integrated RESTful APIs and AWS services** (S3, CloudFront, Amplify) to enable reliable authentication, listings, and transactions, achieving **99.9% uptime** and supporting up to **1,000 concurrent users** through performance optimizations.
- **Mentored junior developers** on state management, testing, and reusable components, reducing bug density and ensuring consistent code quality across the frontend team.

Web Developer, Wolverine Property Management – Halifax, Canada

January 2024 - March 2024

- Developed and deployed a web-based automation tool with a **Python backend** to parse Excel worksheets, apply property-specific billing rates, and generate consolidated reports, cutting a 5–8 hour monthly task to **under 30 seconds (~95%+ time savings)** and eliminating manual calculation errors.
- Built a **React** frontend tailored for accountants to upload spreadsheets, configure parameters, validate inputs, and download processed reports, streamlining the billing workflow and reducing onboarding/training overhead.
- Implemented robust input validation, error handling, logging, and audit trails in the backend to ensure data accuracy and traceability, improving reporting reliability and freeing accounting staff to focus on strategic financial analysis.

PROJECTS

Real-Time Smart Irrigation Scheduler, Capstone Project (Mobile Application + IoT)

May 2025 – July 2025

- Designed and implemented an **IoT-driven irrigation scheduler** that computes reference evapotranspiration (ET_0) and applies crop stage-specific Kc coefficients to generate per-crop irrigation plans.
- Integrated automatic actuation (NodeMCU → solenoid valves) and **designed a production-ready PCB** to consolidate sensor interfaces, power management, and MCU connections for reliable field deployment.
- Built a resilient stack with a **Flutter** mobile app (Hive for local storage/redundancy) and a **Firebase** backend to ingest live sensor streams (soil moisture, LDR), fetch weather forecasts, and execute decision logic that reconciles ET_0 /Kc schedules with forecast and soil moisture, supporting dynamic rescheduling, offline use, and robust fail-safes.
- Field-tested on lettuce and tomato, delivering **10–35% yield increases**, **~60% reduction in post-planting labor**, and **35–60% water savings**, transforming a multi-hour manual process into an automated, near-real-time workflow.

ProduceChain, Decentralized Produce Tracker (DApp)

March 2025 – April 2025

- Built an end-to-end DApp with **Solidity** smart contracts (EVM) and a **React** frontend, designing on-chain data models, mappings, and access-control modifiers to enforce provenance and an immutable audit trail for every produce item.
- Instrumented granular, timestamped events and wallet-based transaction flows in the UI, enabling off-chain indexing to compute time deltas between nodes and surface bottlenecks or slowdowns across the supply chain.
- Delivered measurable business value by improving traceability and quality assurance, reducing post-harvest loss through faster market connectivity for small/medium farmers, and providing tamper-proof records for dispute resolution and buyer confidence.

Bodfit, Health and Fitness Application

May 2024 – July 2024

- Led the UI/UX design in **Figma** and implemented the workout module in **Flutter**, collaborating with a cross-functional team to deliver polished screens and a responsive in-app experience.
- Developed a dynamic workout engine that generates weekly routines and supports both **body-area** (upper-body, core, lower-body) and **goal-specific** (strength, weight loss, bodybuilding) programs, with per-set scheduling and an integrated in-app timer to guide sessions.
- Integrated an **exercise animation API** for movement demos and used **Firebase** for authentication, user profiles, and personalization (saved progress and tailored workout recommendations), enabling frictionless onboarding and individualized training plans.

Quizzler, Trivia Question Web Game

May 2024 – July 2024

- Built a category- and difficulty-based multiple-choice quiz engine in **React**, integrating the **TriviaDB** API for randomized questions, enforcing per-question timers, lives-based difficulty, real-time scoring, and end-of-game summaries.
- Managed complex game state and network caching for smooth play (reducing redundant API calls), added UX/resilience features (responsive layout, ARIA accessibility), and persisted high scores/progress in **localStorage** for session recovery and replayability.

SKILLS

Backend Development: Python, Node.js, MySQL, MongoDB, PostgreSQL,

Frontend Development: ReactJS, TypeScript, JavaScript, TanStack, ShadcnUI, Bootstrap, Tailwind CSS

Mobile App Development: Flutter, Hive, Firebase

Embedded Systems: Arduino, C/C++, Assembly

Miscellaneous: Docker, AutoCAD, Microsoft Office (Word, Excel, PowerPoint)

HONORS/AWARDS

- 2nd Place Robotics Hackathon, African Leadership University, Rwanda
- Certified Lego Spike Training Coach, Coderina

October 2023

August 2023